

The Magic of Automatic Calculus

Dual Number
Multiplication Rule
 $(a, b) \times (c, d) = (a \times c, a \times d + b \times c)$

They are the same rule!

Calculus Product Rule
 $(f * g)' = f' * g + f * g'$

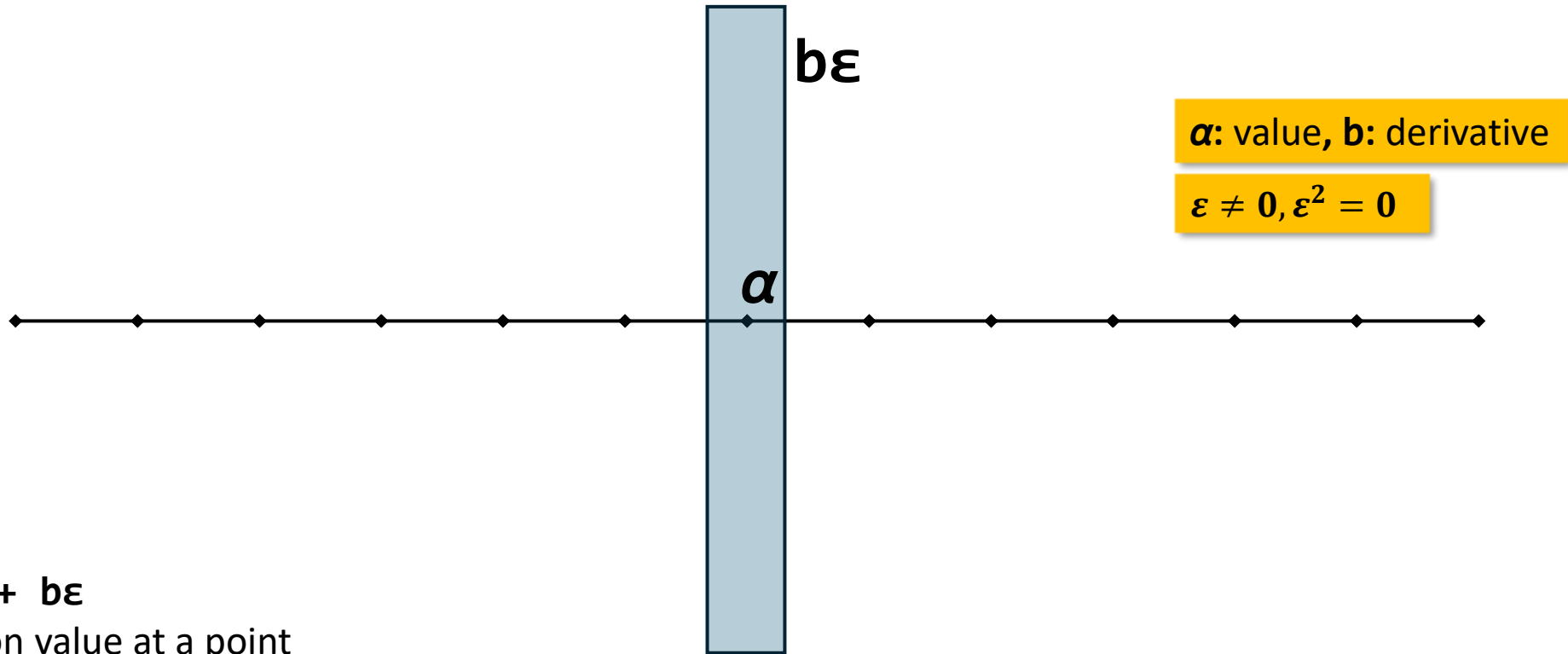
Arithmetic Operations Are Calculus Rules

- Addition: $(a, b) + (c, d) = (a+c, b+d) \rightarrow$ The Sum Rule: $(f+g)' = f' + g'$
- Multiplication: $(a, b) \times (c, d) = (a \times c, a \times d + b \times c) \rightarrow$ The Product Rule: $(f \times g)' = f' \times g + f \times g'$
- The Result: By doing simple arithmetic, we automatically compute exact derivatives.

The Core Idea: Dual Numbers Encode Calculus via a simple “evaluation process”:

- Algebraic form of Dual number: $a + b\epsilon$ (ϵ a special symbol with $\epsilon^2 = 0$)
- Convenient ordered-pair form of Dual number: (a, b)
- **a: the function value, b: the derivative**
- Arithmetic rules mirror calculus rules
- Product rule appears automatically

What Are Dual Numbers?



- Form: $\alpha + b\varepsilon$
- α : function value at a point
- b : derivative at that point
- Defining property: $\varepsilon^2 = 0$ (nilpotent)

***Nilpotent:** a matrix or an algebraic element that, when multiplied by itself a sufficient number of times, results in a zero matrix or zero element, respectively*

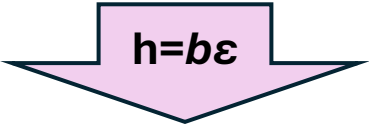
ϵ in Action: First-Order Taylor Expansion

The **Taylor series** of a function is an infinite sum of terms calculated from the values of its derivatives at a single point. It's an essential method for **function approximation**, numerical analysis, and solving differential equations.

Standard Taylor Expansion expansion around point a

(h is the increment from a to x , i.e. $h = x - a$)

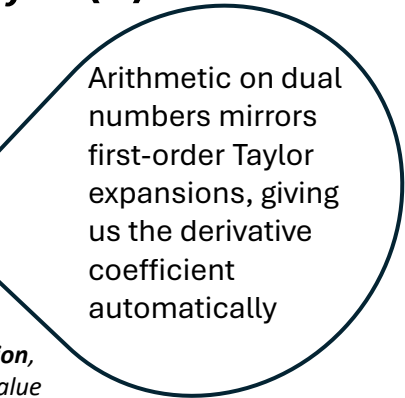
$$f(a + h) = f(a) + f'(a) \cdot h + f''(a) \cdot h^2/2! + f'''(a) \cdot h^3/3! + \dots$$


$$h = b\epsilon$$

$$f(a + b\epsilon) = f(a) + f'(a) \cdot b\epsilon$$

Automatic first-order expansion!

*The **first-order Taylor expansion**, or **linear approximation**, approximates a function $f(x)$ around a point a using its value and its first derivative at that point*



Arithmetic on dual numbers mirrors first-order Taylor expansions, giving us the derivative coefficient automatically